

Neuraxis: Complete Claude Code Experiment and Publication Execution Plan

Executive directive

This document is the controlling specification for continuing, building, testing, executing, auditing, analyzing, and documenting the Neuraxis research project. Claude Code must read the entire document before modifying the repository. It must first inventory and verify work that already exists, reuse valid components, preserve old files and results, and record every conflict between this specification and prior work.

Neuraxis is a computational-neuroscience validation framework. It asks whether a transformed conductance-based neuron model can remain structurally valid, execute successfully, and pass one conventional stimulation test while displaying a scientifically meaningful behavioral change under another stimulation condition. The proposed contribution is a mutation-calibrated perturbation-fingerprint method that selects a compact set of stimulation protocols for detecting behavioral changes missed by canonical regression testing.

The project must not be presented as a Claude Code benchmark. Claude Code may assist software development and may later serve as one secondary transformation source, but the paper's primary contribution must remain the Neuraxis method and the empirical neuroscience/software-reliability result.

The study must proceed through separate development and held-out phases. Claude may autonomously complete infrastructure, model curation, numerical calibration, controlled-variant generation, development execution, and development analysis. It must then stop for human audit. The held-out experiment may begin only after the method is frozen, the prospective evaluation is registered, and the required authorization is provided.

No journal publication is guaranteed. A strong positive result, a narrower result, or a well-designed negative result may support different publication routes, but scientific quality cannot be replaced by more simulation runs. Existing infrastructure already covers NeuroML representation, simulation, scientific testing, and electrophysiological feature extraction; Neuraxis must build on these systems instead of presenting their functions as new.^{[1][2][3]}

Project identity

Project name: Neuraxis

Software title:

Neuraxis: Perturbation-Fingerprint Testing for Behavioral Preservation in Computational Neuron Models

Positive-result manuscript title:

Neutral manuscript title:

Evaluating Perturbation-Fingerprint Tests for Behavioral Preservation in Computational
Neuron Models

The positive title may be used only if the frozen held-out study establishes silent drift. The name Neuraxis remains provisional until Claude documents searches of GitHub, GitLab, PyPI, Google Scholar, PubMed, IEEE Xplore, arXiv, bioRxiv, Zenodo, Google, and relevant trademark databases.

Plain-language explanation

A computerized neuron model contains mathematical descriptions of membrane voltage, ion channels, conductances, capacitance, and related properties. A simulator injects a virtual electrical current into that model and calculates how its voltage changes over time. Large voltage peaks correspond to action potentials, commonly called spikes.

Researchers and software tools routinely modify model files. A file can remain legal, execute without crashing, and produce a familiar response under one standard current injection even though its behavior has changed elsewhere. For example, an original and transformed model might both fire five spikes under the standard test, but only the original might produce rebound firing after a hyperpolarizing input.

Neuraxis therefore tests each model under multiple current-injection protocols. It extracts interpretable features such as spike count, firing frequency, first-spike latency, action-potential width, adaptation, sag, and rebound firing. The measurements across all protocols form the model's **perturbation fingerprint**.

Controlled model modifications are introduced to test whether different validation strategies detect them. Neuraxis then finds a compact set of stimulation protocols that detects many behavior-changing modifications without requiring every possible simulation.

Why the research matters

Conductance-based neuron models are nonlinear. Different parameter combinations may produce similar outputs under restricted observations while responding differently to new inputs or interventions. Consequently, reproduction of one voltage trace is not necessarily strong evidence that a transformation preserved the model's broader behavior.^{[4][5][^6]}

A schema validator can determine whether a NeuroML document follows structural rules, and a simulator can determine whether the model executes. Neither result alone establishes preservation of the model's scientific behavior. Neuraxis addresses the gap between software-level success and tested behavioral preservation.

This matters for model reuse, simulator conversion, refactoring, research reproducibility, and AI-assisted scientific programming. The project could contribute a reusable benchmark, a

catalog of neuron-model failure modes, a compact protocol-selection method, public software, and evidence about when canonical testing is insufficient.

Novelty boundary

Established components

Neuraxis must not claim the following as original:

- NeuroML model representation.
- LEMS model and simulation descriptions.
- pyNeuroML integration.
- jNeuroML validation and execution.
- SciUnit and NeuronUnit scientific testing.
- eFEL electrophysiological-feature extraction.
- Ordinary regression testing.
- Mutation testing.
- Metamorphic testing.
- Greedy maximum coverage.
- Generic test-suite minimization.

Mutation testing inserts controlled modifications and measures whether a test suite detects them. Metamorphic testing checks required relationships among related executions when the exact correct output may be unavailable.^{[7][8]}

Candidate original contribution

The defensible candidate contribution is:

A mutation-calibrated framework that constructs electrophysiological perturbation fingerprints of conductance-based neuron models, identifies transformation-induced behavioral changes missed by a canonical stimulation test, and selects a compact stimulation battery that is evaluated on unseen models and an unseen mutation family.

The literature search must not assert absolute novelty prematurely. Claude must maintain a prior-art matrix containing at least 20 close papers and systems, and the researcher must manually inspect the five closest sources.

Operational definitions

Base model

An original neuron model with verified provenance, documented reuse rights, stable execution, and a recorded immutable hash.

Transformation

Any change to a model or its simulation configuration.

Valid transformation

A representational change intended to preserve behavior, such as a correct unit conversion, complete identifier renaming, or equivalent file reorganization.

Mutant

A deliberately modified version of a base model used to evaluate the detection system.

Mutation family

A category of controlled modifications, such as maximal-conductance scaling, reversal-potential changes, or stimulus-duration errors.

Canonical protocol

The single current-injection protocol representing conventional regression testing.

Candidate protocol battery

The full collection of stimulation protocols considered during development.

Perturbation fingerprint

The collection of trace-level and electrophysiological-feature measurements produced by a model across multiple protocols.

For model m , protocol set P , and feature set E , the fingerprint is:

$$F(m) = \{f(m, p) : p \in P, f \in E\}.$$

Behavioral drift

A reproducible difference between an original and transformed model that exceeds a frozen threshold calibrated using numerical-convergence evidence.

Silent behavioral drift

A transformation that:

1. Passes structural and dimensional checks.
2. Executes successfully.
3. Passes the canonical regression protocol.
4. Displays behavioral drift under at least one other frozen protocol.

Empirical reference battery

The complete frozen candidate battery used to establish whether a transformation has an observable effect within the tested domain. It is not proof of universal inequivalence.

Admissible non-equivalent mutant

A single-fault mutant that executes, has complete provenance, and displays a reproducible difference under the empirical reference battery beyond calibrated numerical uncertainty.

Empirical behavioral certificate

A statement that a model passed the finite Neuraxis battery. It must never be described as formal proof of equivalence, complete biological validation, or universal safety.

Research questions

Primary question

On neuron models excluded from method development, does a compact development-selected perturbation battery detect more admissible non-equivalent transformations than a single canonical regression protocol?

Secondary questions

- How many variants pass structural checks and execute but differ behaviorally?
- How many non-equivalent variants survive canonical regression?
- Which protocols expose the most silent drift?
- Does a selected battery outperform cost-matched random batteries?
- How many protocols are needed before coverage gains diminish?
- Which features detect different mutation families?
- Does performance generalize to completely unseen models?
- Does performance generalize to a held-out mutation family?
- What false-positive rate occurs on valid transformations?
- Are conclusions robust to alternative tolerances?

- Can Neuraxis detect behavioral changes in AI-assisted model transformations?

Hypotheses

Primary confirmatory hypothesis

The frozen Neuraxis battery will detect a higher proportion of admissible non-equivalent mutants than canonical regression on held-out neuron models.

Random-baseline hypothesis

At the same protocol count or runtime budget, Neuraxis will outperform the frozen distribution of randomly selected protocol batteries.

Model-generalization hypothesis

A battery selected on development models will retain useful detection performance on entirely unseen models.

Mutation-family hypothesis

The battery will detect at least some non-equivalent mutants from a family excluded during protocol selection.

Valid-transformation objective

Neuraxis should maintain a low false-positive rate on verified valid transformations. No target rate may be invented after held-out results are visible.

Agent hypothesis

Some transformations produced by the evaluated Claude Code configuration may pass basic checks while failing hidden Neuraxis checks. This hypothesis is secondary; the main paper must survive if Claude makes no errors.

Existing-repository continuation directive

The repository already contains partial work. Claude must not assume a clean start.

Before implementing anything, Claude must:

1. Read this entire specification.
2. Inspect every branch, commit, tag, file, directory, model, configuration, result, manifest, audit sheet, and planning document.
3. Identify completed, partial, unverified, obsolete, and conflicting work.
4. Locate all NeuroSem, PerturbPrint, and Neuraxis references.
5. Preserve historical names and paths when changing them would damage provenance.

6. Determine whether any raw data were overwritten.
7. Determine whether any held-out outcomes were previously accessed.
8. Determine whether existing results are exploratory, confirmatory, or status-uncertain.
9. Reuse verified components.
10. Repair invalid components instead of rebuilding blindly.
11. Assign new run IDs to every legitimate rerun.
12. Record all conflicts and resolutions in docs/DEVIATION_LOG.md.
13. Never relabel previously inspected data as untouched confirmation data.
14. If leakage occurred, reclassify the affected split as development and construct a genuinely untouched evaluation split if enough models remain.

The first required document is docs/CURRENT_REPOSITORY_STATE.md, explaining what exists, what is verified, what can be reused, what must be repaired, what data have been inspected, and what decisions require human approval.

Required repository documents

Claude must create or update:

- docs/CURRENT_REPOSITORY_STATE.md
- docs/SPECIFICATION_GAP_ANALYSIS.md
- docs/PRIOR_ART_MATRIX.md
- docs/DEPENDENCY_AUDIT.md
- docs/LICENSING_MATRIX.md
- docs/NAME_AUDIT.md
- docs/PLAIN_LANGUAGE_OVERVIEW.md
- docs/DEVELOPMENT_PROTOCOL.md
- docs/PROTOCOL_CATALOG.md
- docs/FEATURE_CATALOG.md
- docs/MUTATION_CATALOG.md
- docs/STATISTICAL_ANALYSIS_PLAN.md
- docs/PREREGISTRATION_DRAFT.md
- docs/DEVIATION_LOG.md
- docs/DECISIONS_REQUIRED.md
- docs/AI_DISCLOSURE.md
- AI_USE_LOG.md
- CHANGELOG.md

Prior-art audit

The matrix must contain at least 20 close publications or software systems. Each row must record:

- Exact title.
- Authors.
- Year.
- DOI.
- Publisher or archival URL.
- Whether the full text was inspected.
- Models or scientific systems studied.
- Transformation testing.
- Controlled mutation testing.
- Multiple stimulation protocols.
- Electrophysiological features.
- Protocol/test-suite optimization.
- Held-out generalization.
- Coding-agent involvement.
- Overlap with Neuraxis.
- Remaining distinction.

For the five closest works, create a verification packet containing relevant page numbers, brief original paraphrases, and exact overlap. Search-result snippets and AI summaries are not enough.

Recommended repository architecture

```
neuraxis/  
├─ README.md  
├─ LICENSE  
├─ CITATION.cff  
├─ pyproject.toml  
├─ environment.yml  
├─ Dockerfile  
├─ Makefile  
├─ CHANGELOG.md  
├─ AI_USE_LOG.md  
├─ configs/  
│   ├── development.yaml  
│   ├── heldout_frozen.yaml  
│   ├── protocols.yaml  
│   ├── features.yaml  
│   ├── mutations.yaml  
│   ├── tolerances.yaml  
│   └── selection.yaml
```

```
| | | statistics.yaml
| | | agent.yaml
| | docs/
| | manifests/
| | | models.csv
| | | protocols.csv
| | | mutations.csv
| | | transformations.csv
| | | development_split.csv
| | | heldout_split.csv
| | | hashes/
| | models/
| | | raw/
| | | curated/
| | | fixtures/
| | neuraxis/
| | | cli.py
| | | configuration.py
| | | provenance.py
| | | simulators/
| | | protocols/
| | | features/
| | | mutations/
| | | transformations/
| | | validation/
| | | selection/
| | | orchestration/
| | | analysis/
| | tests/
| | | unit/
| | | integration/
| | | regression/
| | | fixtures/
| | workflows/
| | | curate_models.py
| | | run_references.py
| | | calibrate_tolerances.py
| | | generate_variants.py
| | | run_development.py
| | | select_protocols.py
| | | freeze_study.py
| | | run_heldout.py
| | | lock_primary_results.py
| | | run_robustness.py
| | | run_agent_study.py
| | | reproduce_manuscript.py
| | results/
| | | raw/
| | | processed/
| | | audits/
| | | tables/
| | | figures/
| | manuscript/
| | | manuscript.md
| | | supplement.md
| | | cover_letter.md
```

The existing repository may retain an alternative structure if Claude documents why it provides equivalent separation, provenance, and reproducibility.

Environment and dependencies

The reproducible environment should include verified compatible versions of:

- Python.
- Java.
- NeuroML and LEMS tooling.
- pyNeuroML.
- jNeuroML.
- eFEL.
- NumPy.
- SciPy.
- pandas.
- A plotting library.
- Statistical packages.
- Pytest.
- Type checking and linting.
- Docker or another clean-environment mechanism.

Claude must verify actual current APIs and versions from official documentation. It must never implement an assumed function solely because an earlier AI-generated plan mentioned it.

Infrastructure smoke test

Before large-scale collection, demonstrate from a clean environment that the pipeline can:

1. Load two small fixture models.
2. Validate both models.
3. Execute current-clamp simulations.
4. Save complete voltage traces.
5. Extract spike count, first-spike latency, and mean firing frequency.
6. Generate stimulus-and-voltage diagnostic plots.
7. Repeat the run and reproduce outputs within numerical tolerance.
8. Produce complete provenance records.
9. Run automated unit and integration tests.

Failure blocks scaled data collection.

Provenance and immutability

Every execution must receive a unique run ID and a structured record containing at least:

```
{
  "run_id": "unique-id",
  "project_name": "Neuraxis",
  "study_phase": "development_or_heldout",
  "model_id": "model-id",
  "model_hash": "sha256",
  "mutation_id": null,
  "transformation_id": null,
  "protocol_id": "protocol-id",
  "simulator": "name",
  "simulator_version": "version",
  "time_step": 0.025,
  "configuration_hash": "sha256",
  "git_commit": "commit",
  "environment_digest": "digest",
  "start_time": "timestamp",
  "end_time": "timestamp",
  "status": "success_or_failure",
  "runtime_seconds": 0.0,
  "stdout_path": "path",
  "stderr_path": "path",
  "trace_path": "path",
  "trace_hash": "sha256",
  "feature_path": "path",
  "feature_hash": "sha256"
}
```

Raw outputs must never be overwritten. A rerun creates a new run ID and preserves the earlier record. Completed jobs may be skipped only after the orchestrator verifies expected hashes and completeness.

Model curation

Inclusion criteria

A model may enter the study only if it:

- Has identifiable scientific provenance.
- Has documented reuse rights.
- Passes appropriate NeuroML validation.
- Executes without manual intervention.
- Produces an interpretable membrane-voltage trace.
- Has a clear recording location.
- Responds meaningfully to current injection.
- Has available dependencies.

- Supports several relevant mutation operators.
- Completes protocols within the compute budget.
- Produces stable outputs under numerical refinement.

Exclusion criteria

Exclude and log a model if:

- Source or licensing is unclear.
- Required files are missing.
- The reference model fails before mutation.
- Execution is nondeterministic without a scientifically explained reason.
- Recording location is ambiguous.
- Feature extraction is unreliable.
- Output does not stabilize under refinement.
- Runtime makes the fixed design infeasible.
- Ordinary execution requires undocumented manual repair.
- The planned protocols are incompatible.

Exclusion cannot depend on whether a model strengthens the desired conclusion.

Diversity

Prefer a manageable collection spanning several behaviors:

- Tonic firing.
- Spike-frequency adaptation.
- Bursting.
- Rebound firing.
- Sag.
- Different firing thresholds.
- Different channel compositions.

Prioritize simple single-cell models. Include multicompartment models only if stable and feasible.

Licensing plan

Use Apache License 2.0 provisionally for original Neuraxis code, subject to compatibility review.

For every upstream model, record:

- Model ID and title.

- Source publication.
- URL.
- Download date.
- Version or commit.
- SHA-256 hash.
- Original license.
- Redistribution status.
- Attribution requirements.
- Dependencies.

Do not relicense upstream model files. If redistribution is unclear, publish a permitted downloader, source URL, and expected hash rather than bundling

References

1. [\[PDF\] eFEL Documentation](#)
2. [LEMS: a language for expressing complex biological models in ...](#) - Computational models are increasingly important for studying complex neurophysiological systems. As ...
3. [Towards systematic, data-driven validation of a collaborative, multi-scale model of *Caenorhabditis elegans*](#) - The OpenWorm Project is an international open-source collaboration to create a multi-scale model of ...
4. [Multiple models to capture the variability in biological neurons and ...](#) - How tightly tuned are the synaptic and intrinsic properties that give rise to neuron and circuit fun...
5. [Different parameter solutions of a conductance-based model that behave identically are not necessarily degenerate](#)
6. [How Multiple Conductances Determine Electrophysiological ...](#) - Most neurons have large numbers of voltage- and time-dependent currents that contribute to their ele...
7. [Testing convolutional neural network based deep learning systems](#) - Machine learning technology spans many areas and today plays a significant role in addressing a wide...
8. [An Analysis and Survey of the Development of Mutation Testing](#)